Structural Roles of Asymptomatic Infection and a Downstream Bacterial State in a Cholera Model

Myriam Djoukwe Tapi [1,2*] Jules Brice Tchatchueng-Mbougua [2,3]

[1]Department of Mathematics and Computer Science, Faculty of Science, University of Douala, P.O. Box 24157, Douala, Cameroon.
[2]IRD UMI 209 UMMISCO, University of Yaounde I, P.O. Box 337, Yaounde, Cameroon.
[3]Epidemiology and Public Health Department, Centre Pasteur of Cameroon, Yaounde, Cameroon.

Abstract

Additional compartments in cholera models do not necessarily represent additional transmission mechanisms or identifiable biological processes. We analyse an autonomous environmental cholera model with asymptomatic and symptomatic infections, an infectious bacterial reservoir, and a non-infectious downstream bacterial state. After establishing well-posedness, boundedness, the human-generation reproduction number, disease-free stability, endemic equilibrium and forward local bifurcation, we focus on the structural role of the added states. An exact reduction shows that deleting the downstream bacterial state while preserving infectious bacterial removal leaves all upstream transmission trajectories unchanged. A matched hierarchy of models separates this neutral extension from the genuinely transmission-relevant effect of symptom-class allocation. We further exhibit observation-preserving parameter transformations showing that infectious bacterial removal and conversion are indistinguishable when their sum is fixed, and that an unobserved concentration scale introduces an additional ambiguity. Finally, two explicit next-generation decompositions yield reproduction numbers related by a square root while sharing the same invasion threshold. Reproducible simulations verify these structural properties and examine sampled endemic eigenvalues. The contribution is a clarification of state structure, threshold conventions and observation limits within an established class of environmental cholera models; the simulations are illustrative and are not calibrated to surveillance data.

Keywords: cholera; environmental transmission; asymptomatic infection; model reduction; structural identifiability; reproduction number.

1 Introduction

Cholera transmission emerges from interactions between human infection, environmental contamination and the conditions governing exposure to contaminated water. These interactions complicate the relationship between reported disease and the processes sustaining transmission. Environmental models provide a mechanistic interpretation by linking incident infection to a bacterial reservoir and reservoir contamination to infected individuals [1, 2]. A local mathematical model can clarify the consequences of assumptions about infection, recovery and environmental removal before these assumptions are embedded in a spatial or data-driven framework.

Cameroon provides a relevant setting for this research because cholera patterns vary across health districts and climatic subzones. Ngwa et al. documented substantial spatial and seasonal heterogeneity during 2000–2012, including differences in the relationships between reported disease, rainfall, temperature and the built environment [15]. More recently, Hameni Nkwayep et al. studied a seasonally varying model with undetected infections and data assimilation using Cameroonian surveillance observations [16]. These studies establish a regional motivation for transmission modelling, while showing that a homogeneous model cannot by itself explain the country's full epidemiological diversity. The present study is a theoretical contribution motivated by this setting; its numerical illustrations do not estimate transmission parameters from Cameroonian case data.

Environmental cholera models have developed along several complementary lines. Codeço examined the interaction between human infection and the aquatic reservoir in endemic and epidemic dynamics

*Corresponding author: xxxxx@gmail.com

[1]. Tien and Earn distinguished direct and waterborne transmission and investigated their different timescales [2]. Mukandavire et al. used a fitted model to estimate reproduction numbers during the Zimbabwe epidemic [11]. At a broader mathematical level, Shuai and van den Driessche analysed models incorporating nonlinear incidence, multiple infection stages and multiple pathogen states [12]. Thus, environmental transmission, infection heterogeneity and multiple bacterial compartments already have substantial precedents. A further formulation requires an explicit account of what its states represent and which mathematical conclusions depend on their inclusion.

Clinically inapparent infections are relevant to this distinction between the observed epidemic and the underlying infection process. King et al. showed that inapparent infection and immunity assumptions could substantially alter the interpretation of long-term cholera dynamics [8]. Their work supports distinguishing symptom classes when differences in recovery or environmental contamination have a clear epidemiological interpretation. In the present model, a fraction θ of newly infected individuals becomes symptomatic, while the remaining fraction $1-\theta$ enters the asymptomatic class. Their representation makes the contributions of the two classes to the reproduction number explicit, without equating asymptomatic infection with a separately measured surveillance category.

The interpretation of bacterial states requires equal care. Evidence of host-associated changes in infectivity motivated subsequent models with distinct environmental infectivity classes [9, 10]. These models are not equivalent to every formulation containing two bacterial compartments. Here, B_v contributes to infection, whereas B_{nv} is a non-infectious downstream state produced by conversion from B_v. The model contains no return from B_{nv} to B_v. The second state therefore records an environmental quantity without representing bacterial reactivation or an independent source of recurrent infection. This distinction is essential to interpreting the threshold and stability results.

The mathematical question is how saturating environmental exposure and two human infection classes determine invasion, equilibrium existence and local behaviour near the threshold. The next-generation framework provides a consistent definition of the reproduction number [3, 4, 5], while centre-manifold analysis characterizes the local bifurcation [7]. These questions differ from empirical identification: a well-defined threshold does not imply that all its constituent parameters can be estimated from reported cases. Studies of waterborne models have demonstrated limitations in separating environmental parameters from clinical time series and in distinguishing alternative transmission structures [13, 14].

The present contribution is not the introduction of additional compartments per se. Rather, it is a structural clarification of what those compartments do, and do not do, in a local environmental cholera model. The analysis is organized around three questions.

First, which added states modify transmission dynamics and which merely record downstream quantities? We show that the non-infectious bacterial state is dynamically downstream: if the loss from the infectious reservoir is preserved, removing this state leaves all upstream transmission trajectories unchanged.

Second, which parameter combinations are distinguishable from idealized observations? We prove two explicit observation-preserving transformations. The first shows that infectious bacterial removal and conversion cannot be separated from human observations, nor from observations of the infectious reservoir alone, when their sum is fixed. The second shows that an unobserved environmental concentration scale creates a further ambiguity involving bacterial concentration, half-saturation and shedding coefficients.

Third, how do generation conventions affect the numerical value of the reproduction number? We compare two explicit next-generation decompositions and show that they yield reproduction numbers related by a square root while preserving the same invasion threshold.

Thus, the manuscript should be read as a clarification of state structure, threshold conventions and observation limits within an established class of environmental cholera models. Well-posedness, equilibrium and local bifurcation results provide the analytical foundation; numerical illustrations check the stated invariances rather than establish empirical validity. These results are specific clarifications within a known modelling framework, not a claim that environmental compartments, symptom classes or identifiability analysis are themselves new.

Section 2 presents the formulation and Section 3 establishes its mathematical and structural properties. Section 4 treats local sensitivity. Section 5 presents numerical illustrations, followed by discussion and conclusions in Sections 6 and 7. Appendix A details the centre-manifold calculation.

2 Model Formulation

2.1 Model description and assumptions

We consider a homogeneous human population interacting with an aquatic environment that serves as a reservoir for *Vibrio cholerae*. The total human population at time t is given by

$$N(t) = S(t) + I_A(t) + I_S(t) + R(t),$$

where $S(t)$ denotes susceptible individuals, $I_A(t)$ asymptomatic infectious individuals, $I_S(t)$ symptomatic infectious individuals, and $R(t)$ recovered individuals with no return to susceptibility during the modelled period.

The environmental reservoir is described by two bacterial compartments:

$$B_v(t) \quad \text{and} \quad B_{nv}(t),$$

representing an infectious environmental state and a non-infectious downstream state, respectively. The subscripts are retained from the original formulation; they do not identify viability, culturability or a measured biological phenotype.

To describe the main epidemiological and environmental processes involved in cholera transmission, we make the following assumptions:

(A1) New susceptible individuals enter the population at a constant rate Λ, representing births and immigration and experience the natural mortality μ.

(A2) Infection occurs when susceptible individuals are exposed to virulent bacteria in the aquatic environment. We assume a saturating Holling type II functional response, so that the force of infection is given by

$$\beta \frac{B_v}{K + B_v},$$

where $\beta > 0$ is the effective transmission rate and $K > 0$ is the half-saturation constant.

(A3) Following infection, a proportion $\theta \in (0, 1)$ of individuals develops clinical symptoms, while the remaining proportion $1 - \theta$ experiences an asymptomatic infection.

(A4) Asymptomatic individuals recover at rate γ_A and experience natural mortality at rate μ.

(A5) Symptomatic individuals recover at rate γ_S, experience natural mortality at rate μ, and may also die from cholera at an additional rate δ.

(A6) Recovered individuals leave the population through natural mortality at rate μ. Loss of immunity and subsequent return to the susceptible class are not considered over the time scale of the model.

(A7) Both asymptomatic and symptomatic individuals contribute to environmental contamination by shedding bacteria at rates ξ_A and ξ_S, respectively.

(A8) Bacteria in B_v leave the infectious reservoir through removal at rate η or conversion into the non-infectious state at rate ρ. Only their combined rate enters the B_v loss term.

(A9) The downstream state B_{nv} is removed at rate η_{nv}, with no return to B_v and no contribution to infection.

(A10) All parameters are positive constants, with $0 < \theta < 1$. We study the autonomous model with constant β.

The common natural mortality rate μ applies to all human classes; δ represents additional disease-induced mortality only in I_S. Differences in total mortality arise from disease status, not from heterogeneous natural mortality. Figure 1 distinguishes human transfers, contributions to the environmental reservoir and the downstream bacterial state.

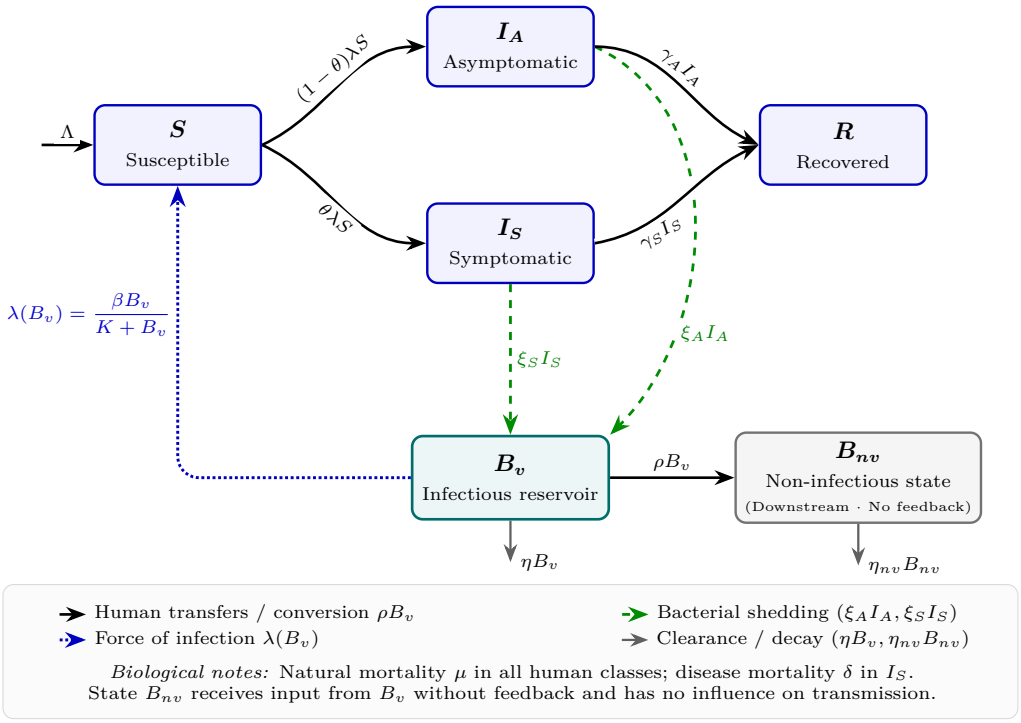


Figure 1: State structure of the local cholera model. Solid black arrows denote human transitions, environmental clearance or bacterial conversion ρ. Dashed green arrows denote shedding contributions from infected humans to the infectious environmental reservoir B_v. The dotted blue arrow represents the catalytic force of infection $\lambda(B_v)$ exerted by the infectious reservoir on susceptible humans. The downstream state B_{nv} receives input from B_v via conversion ρ, but has no feedback to B_v, no direct contribution to infection and no outgoing influence on transmission.

2.2 Model equations

The biological assumptions described above lead to the following system of nonlinear ordinary differential equations, which captures the interaction between cholera transmission in the human population and the dynamics of *Vibrio cholerae* in the aquatic environment:

$$\begin{cases} \dfrac{dS}{dt} = \Lambda - \beta S \dfrac{B_v}{K + B_v} - \mu S, \\ \dfrac{dI_A}{dt} = (1-\theta)\beta S \dfrac{B_v}{K + B_v} - (\mu + \gamma_A) I_A, \\ \dfrac{dI_S}{dt} = \theta \beta S \dfrac{B_v}{K + B_v} - (\mu + \delta + \gamma_S) I_S, \\ \dfrac{dR}{dt} = \gamma_A I_A + \gamma_S I_S - \mu R, \\ \dfrac{dB_v}{dt} = \xi_A I_A + \xi_S I_S - (\eta + \rho) B_v, \\ \dfrac{dB_{nv}}{dt} = \rho B_v - \eta_{nv} B_{nv}. \end{cases} \tag{1}$$

We assume nonnegative initial conditions,

$$S(0),\ I_A(0),\ I_S(0),\ R(0),\ B_v(0),\ B_{nv}(0) \geq 0,$$

so that the model starts from a biologically meaningful state. For simplicity, system (1) is written in vector form as

$$\frac{dX}{dt} = F(X), \qquad X = (S, I_A, I_S, R, B_v, B_{nv})^\top \in \mathbb{R}^6_+.$$

The first four equations describe the evolution of the human population. Susceptible individuals acquire cholera through exposure to virulent bacteria in the aquatic environment. Newly infected individuals enter either the asymptomatic class I_A or the symptomatic class I_S, depending on the fraction θ. Both groups can recover and contribute to bacterial contamination, while symptomatic individuals may also experience disease-induced mortality. The last two equations describe the environmental reservoir. Asymptomatic and symptomatic individuals release virulent bacteria into the aquatic environment at rates ξ_A and ξ_S, respectively. They leave the infectious reservoir through removal or conversion into the non-infectious downstream class. This coupling between human infection and environmental contamination is central to the transmission dynamics captured by the model.

The natural mortality rate μ is common to all human classes; symptomatic individuals additionally experience disease-induced mortality δ. Thus, mortality differs through disease status, but the model does not introduce heterogeneous natural mortality rates. The title and all subsequent calculations follow this choice in the original local formulation.

Here B_v, B_{nv} and K share concentration units, denoted by U. The effective coefficients ξ_A and ξ_S have units U person^{-1}day^{-1} and incorporate the reservoir-volume convention. They are not raw bacterial counts shed per day. The labels "virulent" and "non-virulent" are operational compartment labels: only B_v contributes to the force of infection. No equivalence between non-virulence, non-viability and dormancy is assumed.

Table 1: Parameters and dimensional conventions. Numerical values are specified separately in Table 2.

Parameter	Meaning	Units
Λ	Entry into the susceptible population	persons day^{-1}
μ	Common natural mortality rate	day^{-1}
δ	Additional mortality among symptomatic individuals	day^{-1}
β	Maximum per-susceptible exposure hazard	day^{-1}
θ	Fraction of incident infections that are symptomatic	dimensionless
γ_A, γ_S	Recovery rates	day^{-1}
ξ_A, ξ_S	Effective contributions to environmental concentration	U person^{-1} day^{-1}
η, ρ, η_{nv}	Infectious removal, conversion, non-infectious removal	day^{-1}
K	Half-saturation concentration	U

3 Mathematical analysis

3.1 Basic mathematical properties

We first establish the main mathematical properties of system (1). In particular, we show that the model admits a unique solution, that all state variables remain nonnegative, and that the solutions are bounded for all time. These properties ensure that the model remains biologically meaningful throughout its evolution.

Theorem 1 (Local existence and uniqueness)**.** *For every initial condition $X(0) \in \mathbb{R}^6_+$, system (1) admits a unique local solution*

$$X(t) \in C^1\big([0, T_{\max}), \mathbb{R}^6_+\big).$$

Proof. Let $F : \mathbb{R}^6_+ \to \mathbb{R}^6$ denote the vector field defined by system (1). Since $K > 0$, the function

$$B_v \longmapsto \frac{B_v}{K + B_v}$$

is continuously differentiable for $B_v \geq 0$. Hence, F is continuous and locally Lipschitz on an open neighbourhood of each point of $\mathbb{R}^6_+$ contained in $\{B_v > -K\}$. The Cauchy-Lipschitz theorem therefore guarantees the existence and uniqueness of a local solution. □

The above result guarantees that, for a given initial state and fixed parameter values, the trajectory of the model is uniquely determined.

Lemma 1 (Positivity)**.** *If $X(0) \in \mathbb{R}^6_+$, then $X(t) \in \mathbb{R}^6_+, \quad t \geq 0$. Thus, the nonnegative orthant is positively invariant under system (1).*

Proof. It is sufficient to verify that the vector field points inward on each boundary of the nonnegative orthant. For the susceptible class, if $S = 0$, then

$$\left.\frac{dS}{dt}\right|_{S=0} = \Lambda \geq 0.$$

Similarly, when $I_A = 0$, $I_S = 0$, or $R = 0$, we obtain

$$\left.\frac{dI_A}{dt}\right|_{I_A=0} = (1-\theta)\beta S \frac{B_v}{K + B_v} \geq 0,$$

$$\left.\frac{dI_S}{dt}\right|_{I_S=0} = \theta\beta S \frac{B_v}{K + B_v} \geq 0,$$

and

$$\left.\frac{dR}{dt}\right|_{R=0} = \gamma_A I_A + \gamma_S I_S \geq 0.$$

For the environmental variables,

$$\left.\frac{dB_v}{dt}\right|_{B_v=0} = \xi_A I_A + \xi_S I_S \geq 0,$$

and

$$\left.\frac{dB_{nv}}{dt}\right|_{B_{nv}=0} = \rho B_v \geq 0.$$

Therefore, none of the state variables can become negative, and $\mathbb{R}^6_+$ is positively invariant. □

We next establish the boundedness of the human and bacterial populations.

Theorem 2 (Boundedness)**.** *Every solution of system (1) with nonnegative initial conditions remains bounded for all $t \geq 0$.*

Proof. Let $N(t) = S(t) + I_A(t) + I_S(t) + R(t)$ denote the total human population. Adding the first four equations of system (1) gives

$$\frac{dN}{dt} = \Lambda - \mu N - \delta I_S.$$

Since $I_S \geq 0$,

$$\frac{dN}{dt} \leq \Lambda - \mu N.$$

By the comparison principle,

$$N(t) \leq N(0)e^{-\mu t} + \frac{\Lambda}{\mu}\left(1 - e^{-\mu t}\right),$$

and hence

$$N(t) \leq N_{\max} := \max\left\{N(0), \frac{\Lambda}{\mu}\right\}, \qquad t \geq 0.$$

Since $I_A, I_S \leq N_{\max}$, the equation for B_v gives

$$\frac{dB_v}{dt} \leq (\xi_A + \xi_S)N_{\max} - (\eta + \rho)B_v.$$

Therefore,

$$B_v(t) \leq B_{v,\max} := \max\left\{B_v(0), \frac{(\xi_A + \xi_S)N_{\max}}{\eta + \rho}\right\}.$$

Finally,

$$\frac{dB_{nv}}{dt} \leq \rho B_{v,\max} - \eta_{nv} B_{nv},$$

which yields

$$B_{nv}(t) \leq B_{nv,\max} := \max\left\{B_{nv}(0), \frac{\rho B_{v,\max}}{\eta_{nv}}\right\}.$$

Thus, all components of $X(t)$ remain bounded for $t \geq 0$. □

Corollary 1 (Global existence)**.** *For every nonnegative initial condition, the solution of system* (1) *exists for all* $t \geq 0$ *and satisfies*

$$X(t) \in C^1\big([0,\infty), \mathbb{R}^6_+\big).$$

Proof. By Theorem 1, a unique local solution exists. Lemma 1 and Theorem 2 show that this solution remains in a bounded subset of $\mathbb{R}^6_+$. Consequently, no finite-time blow-up can occur, and the local solution extends to all $t \geq 0$. □

The previous results allow us to define a biologically meaningful positively invariant region. Let

$$N_{\max} = \max\left\{N(0), \frac{\Lambda}{\mu}\right\},$$

$$B_{v,\max} = \max\left\{B_v(0), \frac{(\xi_A + \xi_S)N_{\max}}{\eta + \rho}\right\},$$

and

$$B_{nv,\max} = \max\left\{B_{nv}(0), \frac{\rho B_{v,\max}}{\eta_{nv}}\right\}.$$

Then

$$\Omega = \left\{X \in \mathbb{R}^6_+ : N \leq N_{\max},\ B_v \leq B_{v,\max},\ B_{nv} \leq B_{nv,\max}\right\}$$

is positively invariant. Hence, every trajectory starting in Ω remains nonnegative and bounded for all $t \geq 0$.

3.2 Disease-free equilibrium and reproduction number

Set

$$a_A = \mu + \gamma_A, \qquad a_S = \mu + \delta + \gamma_S, \qquad c = \eta + \rho, \qquad S^0 = \Lambda/\mu,$$

and define

$$Q = \frac{(1-\theta)\xi_A}{a_A} + \frac{\theta\xi_S}{a_S}. \tag{2}$$

The disease-free equilibrium is $E_0 = (S^0, 0, 0, 0, 0, 0)$. For the infected subsystem $Y = (I_A, I_S, B_v)^\top$, count new human infections in $\mathcal{F}$ and environmental production in the transfer terms $\mathcal{V}$. The linearized matrices are

$$F = \begin{pmatrix} 0 & 0 & (1-\theta)q \\ 0 & 0 & \theta q \\ 0 & 0 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} a_A & 0 & 0 \\ 0 & a_S & 0 \\ -\xi_A & -\xi_S & c \end{pmatrix}, \qquad q = \frac{\beta S^0}{K}. \tag{3}$$

Since V is nonsingular and $V^{-1} \geq 0$, the next-generation construction [4] gives

$$\mathcal{R}_0 = \rho(FV^{-1}) = \frac{\beta\Lambda Q}{\mu K c}. \tag{4}$$

Indeed, FV^{-1} has rank one and its only nonzero eigenvalue is qQ/c. This definition counts a complete human–environment–human generation.

Proposition 1 (Comparison of generation conventions)**.** *For the infected coordinates (I_A, I_S, B_v), an alternative decomposition that counts both new human infections and production of infectious environmental bacteria is*

$$F_{\text{split}} = \begin{pmatrix} 0 & 0 & (1-\theta)q \\ 0 & 0 & \theta q \\ \xi_A & \xi_S & 0 \end{pmatrix}, \qquad V_{\text{split}} = \operatorname{diag}(a_A, a_S, c).$$

Its next-generation matrix has characteristic polynomial

$$z\left(z^2 - \frac{qQ}{c}\right),$$

and therefore

$$\mathcal{R}_0^{\text{split}} = \rho(F_{\text{split}} V_{\text{split}}^{-1}) = \sqrt{\mathcal{R}_0^{\text{human}}}, \qquad \mathcal{R}_0^{\text{human}} = \mathcal{R}_0.$$

Both decompositions give the same linearized infected dynamics and the same threshold at one.

Proof. The only nonzero entries of $F_{\text{split}} V_{\text{split}}^{-1}$ connect the two human coordinates to the bacterial coordinate and vice versa. Expansion of its determinant gives the stated polynomial. Moreover, $F_{\text{split}} - V_{\text{split}} = F - V$. Thus changing the generation convention changes neither the differential equations nor their linearized stability. □

The split quantity counts transitions across both types of infected entity rather than a complete human generation. Its square-root relation holds for this exclusively waterborne structure and these explicit decompositions; it is not a universal identity for models with additional transmission routes. Threshold agreement does not make the two numerical reproduction numbers interchangeable [2, 5].

The contributions $\mathcal{R}_A = \beta\Lambda(1-\theta)\xi_A/(\mu K c a_A)$ and $\mathcal{R}_S = \beta\Lambda\theta\xi_S/(\mu K c a_S)$ sum to $\mathcal{R}_0$. These are model-based pathway contributions rather than proportions of notified cases. The downstream compartments R and B_{nv} add the stable eigenvalues $-\mu$ and $-\eta_{nv}$ and do not change the invasion threshold.

Interpretation. The two reproduction numbers answer different counting questions. The human-generation quantity counts a complete human–environment–human infection cycle. The split quantity counts transitions between infected human compartments and the infected environmental compartment as separate generation steps. This is why the two numerical values differ while their threshold condition is the same. For this model, $R_0^{\text{split}} < 1$ if and only if $R_0^{\text{human}} < 1$, but numerical comparisons across studies must use the same convention.

3.3 Stability of the disease-free equilibrium

Theorem 3 (Local disease-free stability)**.** *The equilibrium E_0 is locally asymptotically stable if $\mathcal{R}_0 < 1$ and unstable if $\mathcal{R}_0 > 1$.*

Proof. The infected Jacobian is $F - V$. The threshold theorem for a nonnegative F and a nonsingular M-matrix V relates the sign of its spectral bound to $\rho(FV^{-1}) - 1$ [4]. The remaining eigenvalues of the full system are $-\mu, -\mu, -\eta_{nv}$, after the variables are ordered in triangular blocks. This proves the result. Linearization alone is inconclusive at equality. □

Theorem 4 (Global disease-free stability in an invariant region)**.** *If $\mathcal{R}_0 \leq 1$, then E_0 is globally asymptotically stable relative to*

$$\Omega_0 = \{X \in \mathbb{R}^6_+ : N \leq S^0\}.$$

Proof. The identity $\dot{N} = \Lambda - \mu N - \delta I_S$ proves positive invariance of Ω_0. Intersect this region with bacterial bounds from the boundedness argument, enlarged to contain the chosen initial condition, to obtain a compact positively invariant set. Define

$$L = \frac{\xi_A}{a_A} I_A + \frac{\xi_S}{a_S} I_S + B_v.$$

Direct differentiation, using the same removal rate in each coefficient and its corresponding equation, gives the exact identity

$$\dot{L} = \left(\frac{\beta S Q}{K + B_v} - c \right) B_v = c \left(\mathcal{R}_0 \frac{S}{S^0} \frac{K}{K + B_v} - 1 \right) B_v. \tag{5}$$

Since $S \leq S^0$, this derivative is nonpositive for $\mathcal{R}_0 \leq 1$ and strictly negative whenever $B_v > 0$, including at $\mathcal{R}_0 = 1$. A trajectory remaining in $\{\dot{L} = 0\}$ must have $B_v = 0$; its bacterial equation then forces $I_A = I_S = 0$. LaSalle's invariance principle implies convergence to this infection-free set. The remaining stable linear equations yield $R \to 0$, $B_{nv} \to 0$ and $S \to S^0$.

For stability, L bounds the three infected variables uniformly for sufficiently small initial infection. The stable equations for R and B_{nv} then bound their responses to those variables. Finally, $S - S^0$ satisfies a stable linear equation driven by the incidence, which is bounded by a constant times B_v in Ω_0. These bounds establish Lyapunov stability relative to Ω_0, completing asymptotic stability. □

This statement specifies its domain. It does not assert global stability for every nonnegative initial condition outside Ω_0.

3.4 Existence and uniqueness of the endemic equilibrium

Theorem 5. *A strictly positive endemic equilibrium exists if and only if $\mathcal{R}_0 > 1$, and it is unique.*

Proof. Let $\phi = \beta B_v/(K + B_v)$. At an endemic equilibrium,

$$S^* = \frac{\Lambda}{\mu + \phi^*}, \quad I_A^* = \frac{(1-\theta)\phi^* S^*}{a_A}, \quad I_S^* = \frac{\theta \phi^* S^*}{a_S}, \quad B_v^* = \frac{Q \phi^* S^*}{c}.$$

Combining the last expression with $B_v^* = K\phi^*/(\beta - \phi^*)$ gives

$$\phi^* = \frac{\beta \Lambda Q - \mu K c}{K c + \Lambda Q} = \frac{\mu K c (\mathcal{R}_0 - 1)}{K c + \Lambda Q}.$$

It is positive precisely when $\mathcal{R}_0 > 1$ and then satisfies $\phi^* < \beta$. All other equilibrium coordinates follow uniquely. In particular,

$$\begin{aligned} S^* &= \frac{cK + Q\Lambda}{Q(\beta + \mu)}, & B_v^* &= \frac{\mu K}{\beta + \mu}(\mathcal{R}_0 - 1), \\ I_A^* &= \frac{(1-\theta) c B_v^*}{a_A Q}, & I_S^* &= \frac{\theta c B_v^*}{a_S Q}, \\ R^* &= \frac{\gamma_A I_A^* + \gamma_S I_S^*}{\mu}, & B_{nv}^* &= \frac{\rho B_v^*}{\eta_{nv}}. \end{aligned} \tag{6}$$

Below or at the threshold there is no positive endemic equilibrium. □

Existence of E^* does not alone establish attraction of all trajectories.

3.5 Forward bifurcation and local endemic stability

Theorem 6. *With β as bifurcation parameter and the remaining parameters fixed, a forward transcritical bifurcation occurs at*

$$\beta_c = \frac{\mu K c}{\Lambda Q}.$$

For $\beta > \beta_c$ sufficiently close to β_c, the endemic equilibrium is locally asymptotically stable.

Proof. Use the closed subsystem (S, I_A, I_S, B_v) and shift S by S^0. At the critical parameter its Jacobian is

$$J_c = \begin{pmatrix} -\mu & 0 & 0 & -c/Q \\ 0 & -a_A & 0 & (1-\theta)c/Q \\ 0 & 0 & -a_S & \theta c/Q \\ 0 & \xi_A & \xi_S & -c \end{pmatrix}.$$

The infected block is an irreducible Metzler matrix with a strictly positive null vector. Its spectral bound is a simple zero eigenvalue and its other eigenvalues have negative real part. The susceptible eigenvalue is $-\mu$. Choose right and left null vectors

$$w = \begin{pmatrix} -c/(\mu Q) \\ (1-\theta)c/(a_A Q) \\ \theta c/(a_S Q) \\ 1 \end{pmatrix}, \qquad v = \frac{1}{H}\begin{pmatrix} 0 \\ \xi_A/a_A \\ \xi_S/a_S \\ 1 \end{pmatrix},$$

where

$$H = 1 + \frac{c}{Q}\left(\frac{(1-\theta)\xi_A}{a_A^2} + \frac{\theta \xi_S}{a_S^2}\right), \qquad v^\top w = 1.$$

Writing the centre-manifold normal form as

$$\dot{u} = Au^2 + B(\beta - \beta_c)u + \text{higher-order terms},$$

we use $A = \frac{1}{2}\sum_{kij} v_k w_i w_j \partial_{ij} f_k$ and $B = \sum_{ki} v_k w_i \partial_{i\beta} f_k$, evaluated at (E_0, β_c). For the incidence $g = \beta S B_v/(K + B_v)$ the relevant derivatives are $g_{SB_v} = \beta_c/K$, $g_{B_vB_v} = -2\beta_c S^0/K^2$ and $g_{B_v\beta} = S^0/K$. Hence

$$A = -\frac{\beta_c}{H}\left(\frac{c}{\mu K} + \frac{QS^0}{K^2}\right) < 0, \qquad B = \frac{QS^0}{KH} > 0. \tag{7}$$

The normal form therefore has a positive stable branch for $\beta > \beta_c$ near the threshold, consistent with the centre-manifold framework in [7]. The downstream eigenvalues $-\mu$ and $-\eta_{nv}$ remain negative, giving the same local stability conclusion for the full system. □

No global endemic stability result for arbitrary $\mathcal{R}_0 > 1$ is claimed.

3.6 Structural role of the second bacterial state

Proposition 2 (Exact downstream reduction)**.** *For identical initial values of (S, I_A, I_S, R, B_v) and identical parameters in their equations, changing $B_{nv}(0)$ or η_{nv} leaves these five trajectories unchanged. Deleting the B_{nv} equation while retaining the loss $-(\eta + \rho)B_v$ has the same property.*

Proof. The first five equations form a closed system, so uniqueness gives the claim. The remaining coordinate is

$$B_{nv}(t) = e^{-\eta_{nv}t}B_{nv}(0) + \rho\int_0^t e^{-\eta_{nv}(t-s)}B_v(s)\,ds.$$

□

Thus, conversion affects transmission through the removal term ρB_v, whereas retention of the downstream state does not create an additional transmission feedback. Removing both that state and the conversion loss changes two features and cannot isolate the effect of observing the second state.

Remark 1 (A hierarchy with explicit matching). Consider three formulations sharing the demographic and exposure parameters Λ, μ, β, K, the symptomatic-class parameters γ_S, δ, ξ_S, and infectious bacterial loss $c = \eta + \rho$:

1. H_1: (S, I_S, R, B_v), with all incident infections entering I_S; this is the limiting case $\theta = 1$, $I_A(0) = 0$.
2. H_2: (S, I_A, I_S, R, B_v), with incidence split according to $(1-\theta, \theta)$ and the same loss c.
3. H_3: the full six-state system, obtained by appending the B_{nv} equation to H_2.

Writing $C = \beta\Lambda/(\mu K c)$, their human-generation thresholds are

$$\mathcal{R}_0^{(1)} = C\frac{\xi_S}{a_S}, \qquad \mathcal{R}_0^{(2)} = \mathcal{R}_0^{(3)} = C\left(\frac{(1-\theta)\xi_A}{a_A} + \frac{\theta\xi_S}{a_S}\right).$$

Consequently,

$$\mathcal{R}_0^{(2)} - \mathcal{R}_0^{(1)} = C(1-\theta)\left(\frac{\xi_A}{a_A} - \frac{\xi_S}{a_S}\right). \tag{8}$$

Including an asymptomatic class therefore changes the threshold unless the two effective infectious contributions coincide; it need not increase it. The equality between H_2 and H_3 extends to all their shared trajectories for matched upstream initial conditions. Setting $\rho = 0$ while deleting B_{nv} is a different comparison: it also changes c and cannot isolate the downstream state's role.

3.7 Identifiability implications

We use structural identifiability in the idealized sense of distinguishability from exact continuous observations under a stated observation map.The observation map considered here includes human states, human incidence, and optionally B_v, but excludes calibrated observations of B_{nv} unless explicitly stated. This differs from precision with finite noisy data [13, 14]. The following transformations establish specific non-identifiabilities; they do not establish that every remaining parameter is identifiable.

Proposition 3 (Indistinguishable removal and conversion)**.** *Let all other parameters and upstream initial conditions be fixed. For any $h \in (-\eta, \rho)$, replace*

$$\eta \longmapsto \eta + h, \qquad \rho \longmapsto \rho - h.$$

Every trajectory of (S, I_A, I_S, R, B_v) remains unchanged. Thus neither any observation of the human states nor any deterministic clinical-incidence observation can distinguish the separate rates η and ρ. Observing B_v in addition does not remove this ambiguity.

Proof. The transformed parameters remain positive and preserve $c = \eta + \rho$. The closed upstream system is identical before and after the transformation; uniqueness establishes equality of its solution. Any observation depending only on that solution, including $\theta\beta S B_v/(K + B_v)$, is therefore identical. The downstream equation generally changes, but is absent from the assumed observation map. □

Interpretation. This result is stronger than saying that η and ρ may be difficult to estimate from noisy data. Under the stated observation map, the two transformed models generate exactly the same upstream trajectory and exactly the same human incidence in the ideal continuous noiseless setting. Hence the separate biological labels "environmental removal" and "conversion to a downstream state" are not recoverable from human observations alone. Even observing B_v on its model scale does not resolve this ambiguity, because B_v depends on η and ρ only through their sum $c = \eta + \rho$ in the upstream subsystem. Additional information on the downstream state or independent experimental knowledge of one of the two rates is required to assign separate values to η and ρ.

Remark 2 (Additional environmental information). If B_v and B_{nv} are both observed on a calibrated concentration scale, and η_{nv} is independently known, the idealized downstream equation gives

$$\rho = \frac{\dot{B}_{nv} + \eta_{nv} B_{nv}}{B_v} \quad \text{where } B_v > 0.$$

If η_{nv} is unknown, separating ρ from η_{nv} requires additional conditions, such as linear independence of the observed functions B_v and B_{nv} over the observation interval. Recovering $\eta = c - \rho$ further requires that c be known or identifiable from other information. These are conditional statements about exact observations, not a claim of reliable derivative estimation from noisy measurements.

Proposition 4 (Unobserved concentration scale)**.** *For any $\kappa > 0$, simultaneously replace*

$$(B_v, B_{nv}, K, \xi_A, \xi_S) \longmapsto (\kappa B_v, \kappa B_{nv}, \kappa K, \kappa\xi_A, \kappa\xi_S),$$

including the environmental initial conditions. All human trajectories and human-incidence observations remain unchanged, as does $\mathcal{R}_0$.

Proof. The ratio $B_v/(K + B_v)$ is invariant. Both environmental equations are multiplied by κ, and the human equations are unchanged. The transformed solution therefore satisfies the transformed system. In (4), Q and K acquire the same factor. □

Interpretation. The scale transformation expresses a common limitation of environmental transmission models: if bacterial concentration is not observed on an independently calibrated scale, then the same human epidemic can be represented with different numerical concentration units. Multiplying B_v, B_{nv}, K, ξ_A and ξ_S by the same positive factor changes the environmental scale but leaves the force of infection, human trajectories and R_0 unchanged. Consequently, estimates of K, ξ_A and ξ_S should not be interpreted as absolute biological quantities unless the environmental concentration scale and initial conditions are externally constrained. What may be identifiable from human data are invariant combinations of parameters, not necessarily the individual environmental parameters themselves.

4 Sensitivity analysis

For a positive parameter p, define the elasticity $\Upsilon_p = (p/\mathcal{R}_0)\partial\mathcal{R}_0/\partial p$. Let

$$w_A = \frac{(1-\theta)\xi_A/a_A}{Q}, \qquad w_S = \frac{\theta\xi_S/a_S}{Q}, \qquad w_A + w_S = 1.$$

Differentiation of (4) yields

$$\begin{aligned}
\Upsilon_\beta &= \Upsilon_\Lambda = 1, & \Upsilon_K &= -1,\\
\Upsilon_{\xi_A} &= w_A, & \Upsilon_{\xi_S} &= w_S,\\
\Upsilon_\eta &= -\eta/c, & \Upsilon_\rho &= -\rho/c,\\
\Upsilon_{\gamma_A} &= -w_A\gamma_A/a_A, & \Upsilon_{\gamma_S} &= -w_S\gamma_S/a_S,\\
\Upsilon_\delta &= -w_S\delta/a_S, & \Upsilon_{\eta_{nv}} &= 0.
\end{aligned}$$

For the symptom fraction and common mortality rate,

$$\Upsilon_\theta = \frac{\theta}{Q}\left(\frac{\xi_S}{a_S} - \frac{\xi_A}{a_A}\right), \tag{9}$$

$$\Upsilon_\mu = -1 - \frac{\mu}{Q}\left(\frac{(1-\theta)\xi_A}{a_A^2} + \frac{\theta\xi_S}{a_S^2}\right). \tag{10}$$

Consequently, symptom-fraction sensitivity need not be positive for all parameter sets, and common-mortality sensitivity is not exactly -1. Mortality elasticities describe model dependence and are not intervention recommendations. These local elasticities also do not measure the effect of a parameter on every finite-time outcome.

5 Numerical illustrations

5.1 Provenance and computational methods

The starting values were taken as illustrative. We retained these values where compatible with (1), used its susceptible mortality value for the common μ, and implemented the symptom split $(1-\theta, \theta)$. These runs are deterministic model experiments, not observations from a Cameroonian study.

The initial condition is $(499990, 5, 5, 0, 100, 0)$ in the state order of (1). Integration uses `deSolve::ode` with LSODA, relative tolerance 10^{-9} and absolute tolerance 10^{-10}, with daily output over 1095 days. A second run uses tolerances 10^{-11} and 10^{-12}. Raw trajectories, parameters, software versions and numerical checks accompany the manuscript. Tiny negative numerical values near zero are retained in the raw output; the logarithmic display uses a floor and does not establish biological extinction.

Table 2: Illustrative baseline values, with common mortality. No empirical estimation is implied.

Parameter	Value	Parameter	Value
Λ	20	μ	0.00004
β	0.45	K	10^5
θ	0.25	δ	0.005
γ_A	1/5	γ_S	1/7
ξ_A	0.1	ξ_S	0.3
η	0.2	ρ	0.05
η_{nv}	0.15		

Symptomatic prevalence is $I_S(t)$, whereas incident symptomatic infections occur at rate $j_S(t) = \theta\beta S(t)B_v(t)/(K + B_v(t))$. We report cumulative incidence as $\int j_S(t)\,dt$, excluding individuals already infected initially, and symptomatic person-time as $\int I_S(t)\,dt$. Both are approximated by the trapezoidal rule on the daily output grid. The peak time is consequently a daily-grid estimate.

5.2 Transient dynamics and threshold comparison

The baseline gives $\mathcal{R}_0 = 7.9383$. To illustrate the proven subthreshold result, we set $\beta = 0.45(0.8/7.9383)$ using the unrounded value in code and retain all other parameters, giving $\mathcal{R}_0 = 0.8$. This is a mathematical threshold experiment rather than an estimated effect of a public-health intervention. The previous script's values $\beta = 0.12$ and $\eta = 0.45$ would give $\mathcal{R}_0 = 1.0584$ under the present equations and are therefore not used to illustrate extinction below one.

In the baseline experiment, symptomatic prevalence reaches approximately 43761 persons at day 25. Daily-grid integration gives approximately 124432 incident symptomatic infections over the plotted interval. In the subthreshold experiment, the initial infected and environmental states still produce a transient peak of approximately 15.58 symptomatic persons at day 8, followed by decay. Its cumulative incident symptomatic infections are approximately 127.3. Noninteger counts reflect deterministic expected quantities. These values characterize the specified illustrative population and initial condition only.

The baseline trajectory exhibits rapid susceptible depletion and a very deep subsequent trough. A finite-time decline with $\mathcal{R}_0 > 1$ does not contradict instability of the disease-free equilibrium: $\mathcal{R}_0$ is evaluated at the fully susceptible demographic equilibrium. Nor does this simulation prove convergence to the endemic equilibrium. Near-zero states fall below reliable relative numerical resolution, and the constant demographic replenishment in this example is slow.

5.3 Structural role of asymptomatic and downstream states

Figure 4 compares the three models defined in Remark 1. Models H_2 and H_3 use identical upstream initial conditions and the same c. Changes to $B_{nv}(0)$ or η_{nv} alter the downstream trajectory but leave the shared trajectories unchanged up to integration error. In the comparison with H_1, all initially infected persons are assigned to I_S, preserving the initial susceptible population and total infected population. This comparison changes symptom allocation as well as infectious-class composition; it is not an intervention contrast or a refitted model comparison.

The same numerical experiment checks the transformation in Proposition 3: two distinct positive pairs (η, ρ) with the same sum yield indistinguishable upstream outputs. A concentration rescaling checks Proposition 4. The parameter settings, trajectories and scaled discrepancies are exported with the reproducibility files. These checks illustrate exact analytical properties and do not quantify practical identifiability from surveillance data. Thus, Figure 4 illustrates the main structural message of the paper: adding a measured downstream state can change the recorded environmental output without changing the transmission subsystem, whereas changing symptom allocation modifies the invasion threshold and symptomatic dynamics.

5.4 Equilibrium branch, sensitivity and stability checks

At the baseline, the human-pathway weights are $w_A = 0.4251$ and $w_S = 0.5749$. The elasticity with respect to θ is 0.4332, while the elasticity with respect to μ is -1.00024. Figure 5 displays numerical central-difference evaluations of the analytical elasticities. Figure 6 plots the exact equilibrium formula as

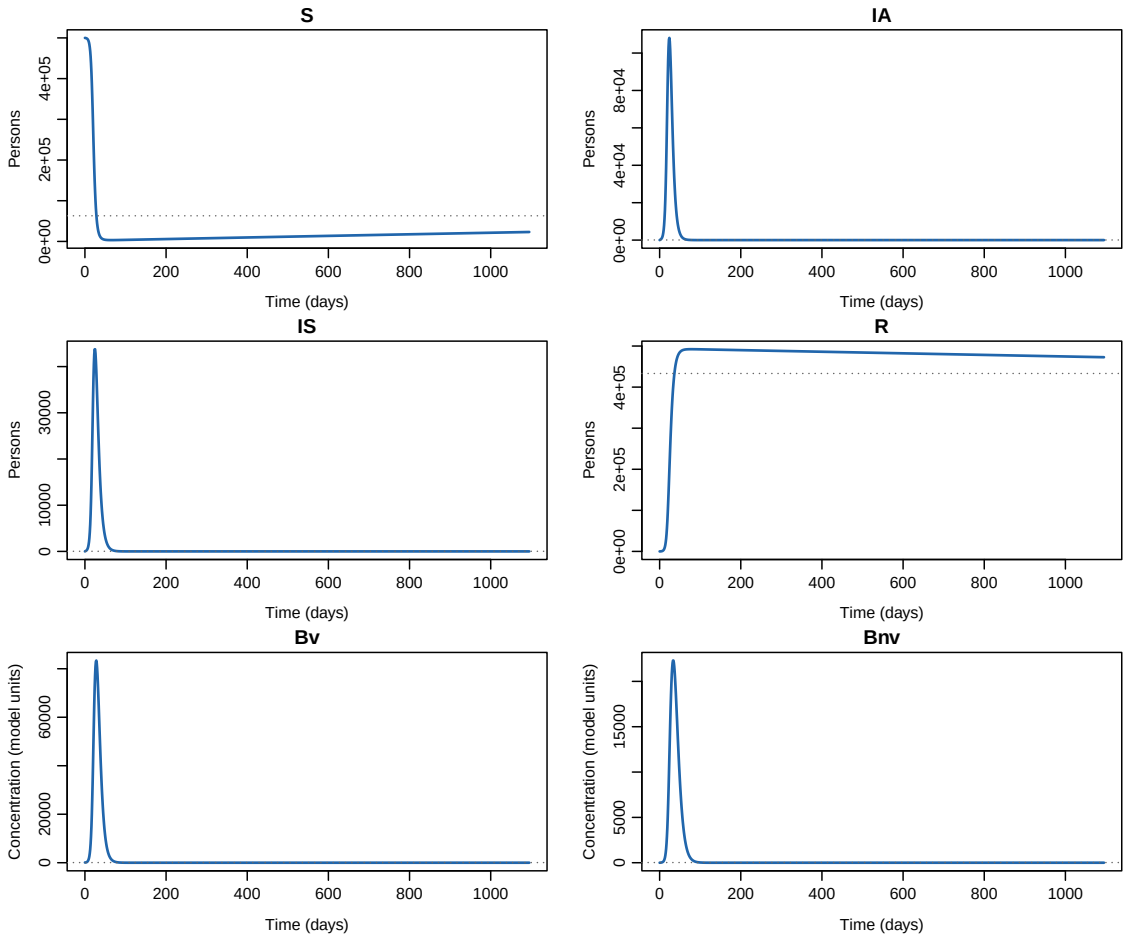


Figure 2: Baseline trajectories for the local model. Dotted lines denote the analytical endemic equilibrium. Their inclusion does not imply that this equilibrium is reached during the plotted interval. Human counts and environmental concentrations have separate panels and units.

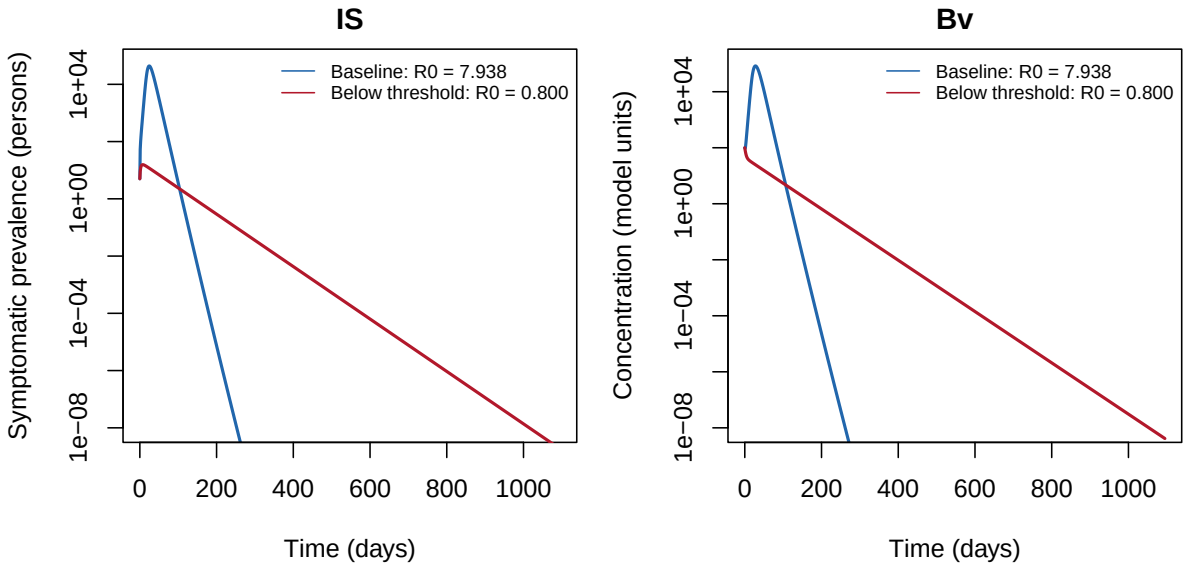


Figure 3: Transient symptomatic prevalence and infectious environmental concentration for the baseline and subthreshold experiments. Logarithmic axes emphasize decline; values below the displayed range are not quantitatively resolved by this figure. In particular, deep numerical troughs are not evidence of exact deterministic extinction.

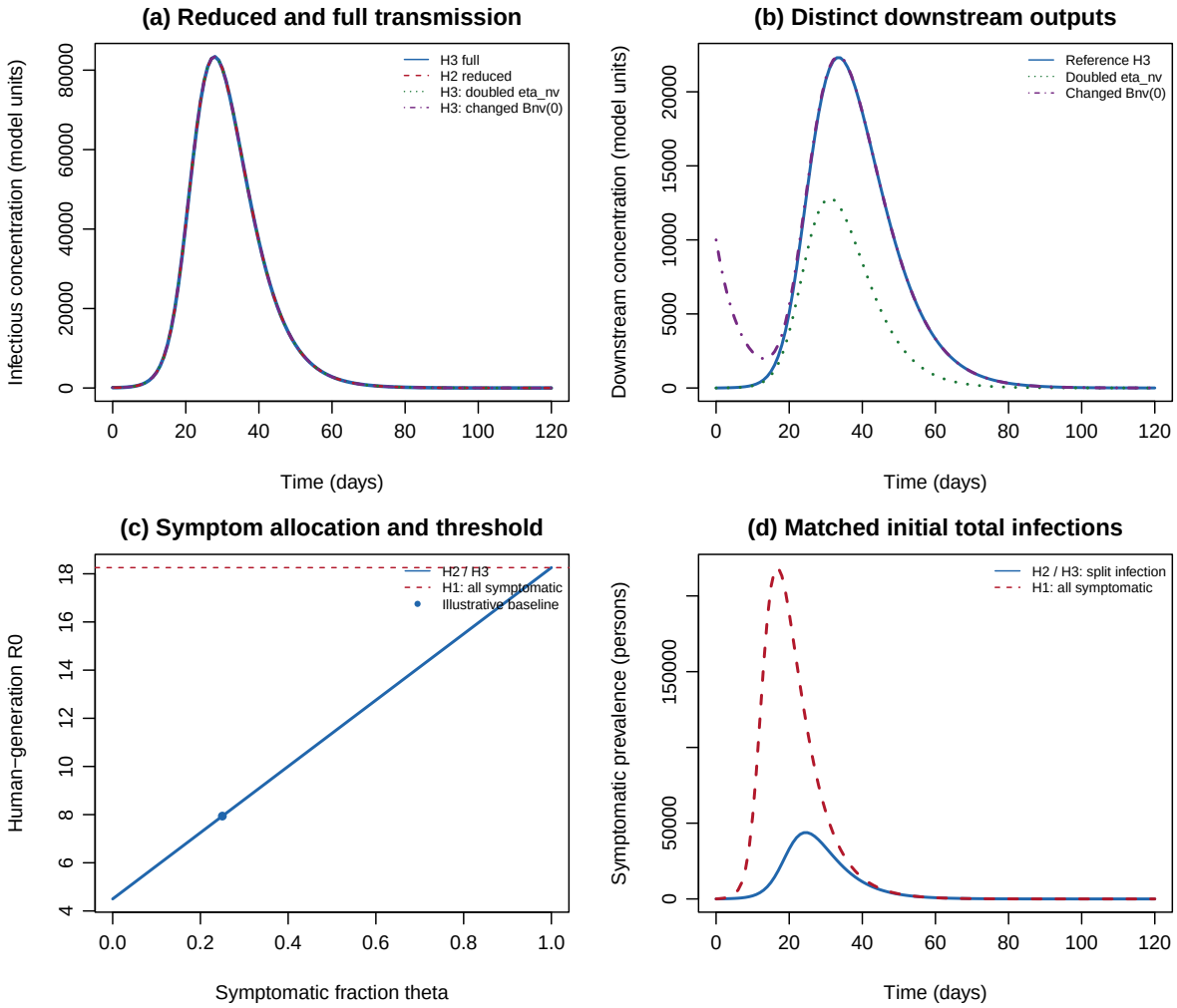


Figure 4: Matched structural comparisons. (a) Infectious reservoir trajectories from the reduced and full systems, including changes to downstream initial conditions and removal. (b) The corresponding downstream trajectories differ. (c) The human-generation threshold varies with symptom allocation; the all-symptomatic model is the endpoint $\theta = 1$. (d) Symptomatic prevalence in the all-symptomatic and split-infection formulations, with identical initial total human and infected populations. Coincident curves in panel (a) express exact decoupling; differences in panel (d) combine changes in symptom allocation and infectious durations. All values are illustrative.

the exposure parameter varies; it illustrates a forward branch, not a simulation-derived proof of global stability. Figure 7 supplements that branch with the largest real part of the analytic Jacobian eigenvalues at a finite set of endemic equilibria. The calculation includes points close to the threshold and a wider illustrative range; it supplies numerical local-stability evidence only for the tested parameter settings.

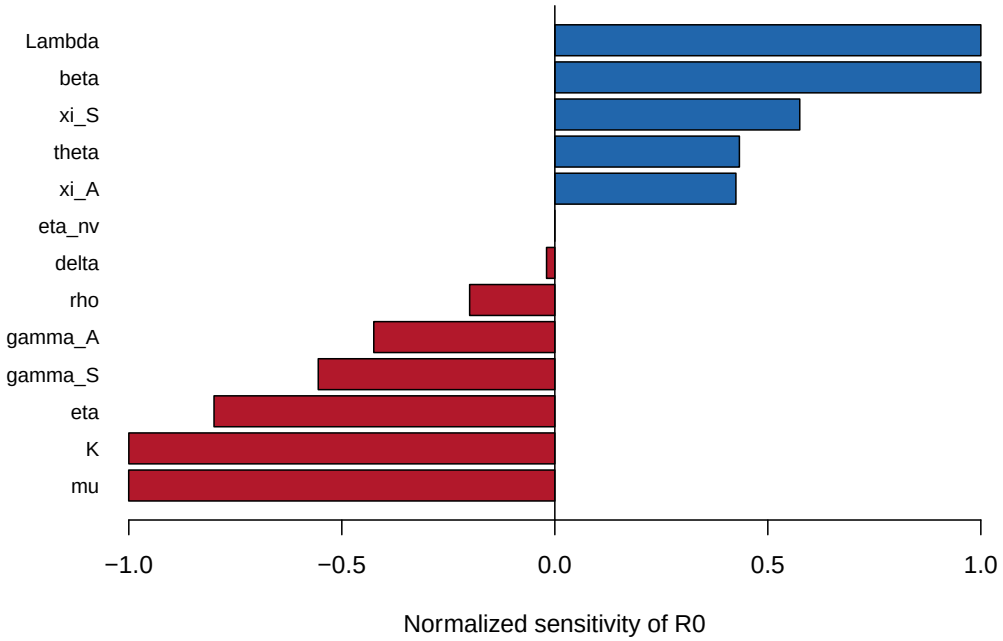


Figure 5: Local elasticities of $\mathcal{R}_0$ at the illustrative baseline. A zero elasticity for η_{nv} follows from the triangular model structure.

5.5 Numerical consistency checks

Substitution of the analytical endemic equilibrium into the six equations gives a maximum absolute residual below 10^{-12} in this example. The spectral radius of the implemented human-generation matrix agrees with (4) within 10^{-12}. The human population-balance identity also holds to numerical precision. Tightening integration tolerances changes the baseline trajectories by less than 3×10^{-9} under the scaled metric $\max_{t,k} |x_{k,\text{tight}} - x_{k,\text{base}}|/(1 + |x_{k,\text{tight}}|)$ on the recorded grid. This absolute-scaled check does not establish relative accuracy in extremely small troughs. Doubling η_{nv} leaves the upstream trajectories unchanged within 5×10^{-9} using the analogous metric. These checks assess implementation consistency rather than empirical validity.

6 Discussion

The analysis distinguishes three properties that are often compressed into a single threshold interpretation: invasion near a disease-free state, existence of a positive endemic equilibrium, and attraction toward that equilibrium. The reproduction number characterizes invasion under the demographic and exposure assumptions of the model. The endemic equilibrium provides a candidate persistent state, while bifurcation analysis supports local stability sufficiently close to the threshold. These statements have different mathematical scopes. Equilibrium existence and finite-time simulation do not establish global endemic stability for every parameter set above the threshold. The domain specified for disease-free global stability is likewise part of the result rather than a dispensable technical qualification [4, 6, 7].

These findings belong to an established body of work on environmentally mediated cholera. General models with differential infectivity already incorporate nonlinear incidence, multiple infection stages and multiple pathogen states, with global threshold results under their corresponding assumptions [12]. The present model should therefore be assessed through the precision of its assumptions, its generation conventions and its observation structure. Unlike the sequential infection stages in Shuai and van den Driessche's formulation, the present incident infections enter two parallel clinical classes. This structural

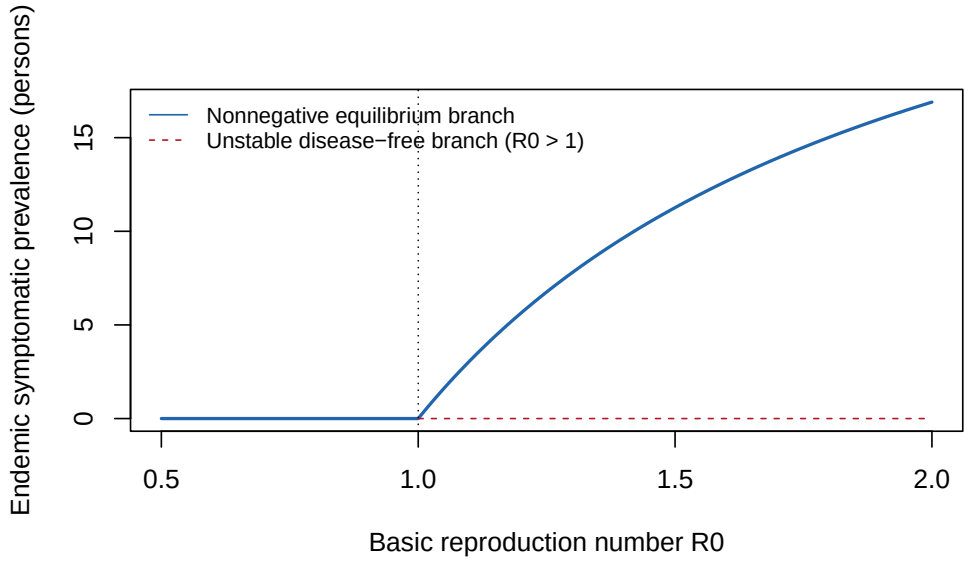


Figure 6: Exact nonnegative equilibrium branch for symptomatic prevalence. The disease-free branch is unstable above one. Endemic stability is proved here only sufficiently close to the threshold; no stability assertion for the entire positive curve follows from this plot.

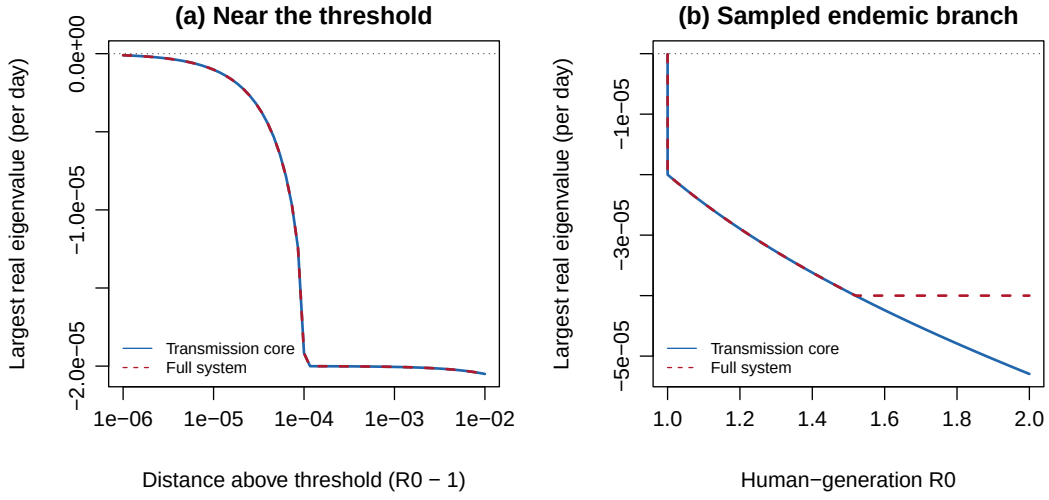


Figure 7: Largest real part of the endemic Jacobian eigenvalues, evaluated analytically and then numerically diagonalized. The four-state transmission core and the full six-state system are distinguished. The full system also has downstream eigenvalues $-\mu$ and $-\eta_{nv}$. The left panel resolves the neighbourhood above the threshold; the right panel displays the wider tested range. A negative value verifies local linear stability at a sampled parameter setting, not global attraction or a theorem for every parameter set.

difference should not be equated automatically with a new threshold principle. The contribution here is the joint treatment of exact downstream reduction, matched model comparisons and observation-preserving parameter transformations within this formulation. Neither the number of compartments nor a saturating incidence term alone establishes originality relative to this literature.

Separating asymptomatic and symptomatic infection reveals how incident-infection fractions, recovery and effective contributions to environmental contamination combine in transmission potential. This interpretation is consistent with the importance of inapparent infection in earlier models [8], although the present study does not infer the asymptomatic fraction from surveillance. The influence of the symptom fraction depends on the relative environmental contributions over the infectious durations of the two classes. A larger symptomatic fraction is therefore not universally associated with greater transmission potential. Equation (8) states the comparison explicitly: replacing an all-symptomatic allocation with two clinical classes changes the threshold according to their relative effective infectious contributions. A difference in compartment count alone has no fixed directional implication. Pathway contributions to the reproduction number should also be distinguished from proportions of reported cases, because reporting is an observation process absent from the model.

The second bacterial compartment has a more restricted role than its inclusion might initially suggest. Proposition 2 and the matched hierarchy show that because B_{nv} neither contributes to infection nor feeds back into B_v, its initial value and removal rate cannot independently alter the upstream transmission equations. Conversion affects transmission through removal from the infectious reservoir; recording the converted bacteria does not create another transmission loop. This structure differs from models in which multiple environmental states retain different degrees of infectivity [10]. It also does not reproduce the full biological processes underlying host-associated changes in infectivity [9]. Non-virulence, non-infectivity, non-viability and dormancy should consequently not be treated as interchangeable descriptions of B_{nv}.

Mortality requires an explicit interpretation. The local equations use a common natural mortality rate and add disease-induced mortality only to symptomatic infection. They distinguish total mortality by disease status without modelling heterogeneous natural mortality between human classes or districts. Mortality-dependent reductions in transmission potential describe removal within the system; they do not imply a favourable health outcome. Public-health interpretation should concern recovery and reduced environmental exposure within the assumed model, while numerical intervention recommendations require evidence that accounts for observed disease burden and uncertainty.

Numerical results are most informative when interpreted alongside these distinctions. A transient decline after an epidemic wave can reflect susceptible depletion even when the reproduction number evaluated at the fully susceptible demographic equilibrium exceeds one. Conversely, initial infected individuals and environmental contamination can produce a temporary rise below the invasion threshold. A finite trajectory therefore need not coincide with a simple endemic-versus-extinction classification. Illustrative simulations can support analytical interpretation and numerical consistency, but do not demonstrate that the same trajectories occur in a particular Cameroonian district. This differs from studies that estimate reproduction numbers or reconstruct epidemic dynamics from observations [11, 16].

Several assumptions limit the epidemiological scope of the formulation. Homogeneous exposure excludes variation in water access, behaviour and contact patterns. Constant recruitment and the absence of return from recovery to susceptibility simplify population and immunity dynamics, especially over long time horizons. The importance of immunity assumptions and alternative transmission mechanisms has been demonstrated in studies of cholera dynamics and model distinguishability [8, 14]. A fixed effective environmental-volume convention is implicit in the contamination coefficients. The two bacterial compartments are therefore effective model states rather than independently established biological measurements.

These assumptions also affect what could be learned from observations. Proposition 3 establishes an exact indistinguishability between infectious bacterial removal and conversion, rather than merely difficulty estimating them. Clinical observations cannot separate these rates even in an ideal continuous noiseless setting, and measuring B_v alone does not break the ambiguity. Remark 2 specifies conditional additional information that could distinguish conversion. Proposition 4 establishes a separate concentration-scale ambiguity when environmental states and initial conditions are unobserved. These structural observations are consistent with evidence that environmental parameters can be difficult to disentangle using outbreak data alone [13]. Accurate fit also does not necessarily identify the correct model structure or guarantee robust extrapolation [14]. The present study establishes mathematical properties under specified assumptions rather than empirical confirmation of every mechanism.

The Cameroonian context provides a rationale for extending the local system while setting limits on its immediate interpretation. Spatial and climatic heterogeneity documented by Ngwa et al. argues against

treating all districts as epidemiologically interchangeable [15]. Spatial models have already shown how water connectivity and human movement can jointly shape regional spread [17, 18]. A metapopulation extension would therefore need to establish the additional consequences of its particular coupling structure and district heterogeneity. Climate-dependent transmission likewise has an established theoretical and empirical literature [19, 20, 16]; a periodic threshold is not generally equivalent to the arithmetic mean of instantaneous autonomous reproduction numbers.

For research supporting public-health decisions, the value of the local analysis is a transparent link between assumptions, threshold quantities and the scope of the conclusions. The model can contribute to a wider programme of spatial and observational research once its parameters, observation process and district-level relevance are evaluated. At this stage, its defensible contribution is theoretical clarification supported by illustrative simulations. Quantitative forecasts, priority-district identification and claims of intervention effectiveness require evidence beyond the properties established here.

7 Conclusion

The local model separates transmission-driving human and environmental states from a non-infectious downstream state. Exact reduction and a matched hierarchy show why adding a tracked compartment need not change the threshold or upstream dynamics. Two observation-preserving transformations distinguish biological parameter labels from information available in human outputs. Explicit next-generation decompositions reconcile the complete human-generation threshold with a square-root split convention. These structural results accompany well-posedness, disease-free stability, endemic equilibrium and local bifurcation analysis. Illustrative simulations support implementation consistency and sampled local-stability checks; they do not establish empirical validity or global endemic attraction. The resulting formulation provides a transparent basis for subsequent spatial and observational research.

Code and data availability

The accompanying scripts `simulations.R` and `structural_checks.R` regenerate all numerical illustrations and export the parameter values, trajectories, model comparisons, eigenvalue calculations and numerical checks used in this manuscript. Running `structural_checks.R` from the project directory also runs `simulations.R`. Software versions are recorded with the validation outputs. These outputs are model-generated illustrations; no individual-level, clinical or surveillance dataset is analysed in this paper. The code is available at https://github.com/jubri84/cholera-structural-model.

Author contributions

J.T. and M.D.T. contributed to model formulation, mathematical analysis, numerical implementation, interpretation of results and manuscript preparation. Both authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Ethics approval

This study develops and analyses a theoretical mathematical model and does not use individual-level, clinical or surveillance data. Ethics approval was therefore not required.

Data availability

No empirical dataset was analysed in this study. All numerical results are generated from the model equations and parameter values reported in the manuscript and accompanying scripts.

A Details of the centre-manifold calculation

The calculation uses the transmission core (S, I_A, I_S, B_v) because the recovered and downstream bacterial states add only stable triangular blocks. Let $x = (S - S^0, I_A, I_S, B_v)$ and $\varepsilon = \beta - \beta_c$. At $(x, \varepsilon) = (0, 0)$, the critical Jacobian and normalized null vectors are those in Section 3. Direct substitution gives $J_c w = 0$, $v^\top J_c = 0$ and $v^\top w = 1$. The infected block is irreducible under the stated strictly positive parameters and $0 < \theta < 1$; its zero eigenvalue is simple and the other eigenvalues have negative real parts.

Let $g(S, B_v, \beta) = \beta S B_v/(K + B_v)$. At the critical point the only nonzero second derivatives involving the state coordinates are

$$g_{SB_v} = g_{B_v S} = \frac{\beta_c}{K}, \qquad g_{B_v B_v} = -\frac{2\beta_c S^0}{K^2}.$$

The mixed parameter derivative needed for the linear unfolding is

$$g_{B_v \beta} = \frac{S^0}{K};$$

$g_{S\beta} = 0$ at $B_v = 0$. The nonlinear incidence enters the first three core equations with coefficients $(-1, 1 - \theta, \theta)$; the bacterial equation is linear. Since $v_S = 0$,

$$v_{I_A}(1 - \theta) + v_{I_S}\theta = \frac{Q}{H}.$$

With the convention $\dot{u} = Au^2 + B\varepsilon u + \cdots$, the ordered mixed state derivatives contribute twice before the factor $1/2$. Therefore

$$\begin{aligned} A &= \frac{Q}{H}\left(g_{SB_v} w_S w_{B_v} + \frac{1}{2} g_{B_v B_v} w_{B_v}^2\right) \\ &= \frac{Q}{H}\left(-\frac{\beta_c c}{\mu Q K} - \frac{\beta_c S^0}{K^2}\right) = -\frac{\beta_c}{H}\left(\frac{c}{\mu K} + \frac{Q S^0}{K^2}\right) < 0, \\ B &= \frac{Q}{H} g_{B_v \beta} w_{B_v} = \frac{Q S^0}{K H} > 0. \end{aligned}$$

Some centre-manifold statements define the quadratic coefficient without the factor $1/2$; that coefficient equals $2A$ and has the same sign [7]. Here the nonzero local branch satisfies $u = -B\varepsilon/A + O(\varepsilon^2)$, which is positive for $\varepsilon > 0$. Its leading centre eigenvalue is $-B\varepsilon$, while all remaining eigenvalues remain negative sufficiently close to the critical point. The triangular downstream blocks retain eigenvalues $-\mu$ and $-\eta_{nv}$, so the full system has the same forward local exchange of stability.

References

[1] Codeço CT. Endemic and epidemic dynamics of cholera: the role of the aquatic reservoir. *BMC Infectious Diseases*. 2001;1:1. doi:10.1186/1471-2334-1-1.

[2] Tien JH, Earn DJD. Multiple transmission pathways and disease dynamics in a waterborne pathogen model. *Bulletin of Mathematical Biology*. 2010;72:1506–1533. doi:10.1007/s11538-010-9507-6.

[3] Diekmann O, Heesterbeek JAP, Metz JAJ. On the definition and the computation of the basic reproduction ratio R_0 in models for infectious diseases in heterogeneous populations. *Journal of Mathematical Biology*. 1990;28:365–382. doi:10.1007/BF00178324.

[4] van den Driessche P, Watmough J. Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. *Mathematical Biosciences*. 2002;180:29–48. doi:10.1016/S0025-5564(02)00108-6.

[5] Diekmann O, Heesterbeek JAP, Roberts MG. The construction of next-generation matrices for compartmental epidemic models. *Journal of the Royal Society Interface*. 2010;7:873–885. doi:10.1098/rsif.2009.0386.

[6] LaSalle JP. The Stability of Dynamical Systems. *Philadelphia: Society for Industrial and Applied Mathematics*. 1976.

[7] Castillo-Chavez C, Song B. Dynamical models of tuberculosis and their applications. *Mathematical Biosciences and Engineering*. 2004;1:361–404. doi:10.3934/mbe.2004.1.361.

[8] King AA, Ionides EL, Pascual M, Bouma MJ. Inapparent infections and cholera dynamics. *Nature*. 2008;454:877–880. doi:10.1038/nature07084.

[9] Merrell DS, Butler SM, Qadri F, et al. Host-induced epidemic spread of the cholera bacterium. *Nature*. 2002;417:642–645. doi:10.1038/nature00778.

[10] Hartley DM, Morris JG Jr, Smith DL. Hyperinfectivity: a critical element in the ability of *V. cholerae* to cause epidemics? *PLoS Medicine*. 2006;3:e7. doi:10.1371/journal.pmed.0030007.

[11] Mukandavire Z, Liao S, Wang J, Gaff H, Smith DL, Morris JG Jr. Estimating the reproductive numbers for the 2008–2009 cholera outbreaks in Zimbabwe. *Proceedings of the National Academy of Sciences USA*. 2011;108:8767–8772. doi:10.1073/pnas.1019712108.

[12] Shuai Z, van den Driessche P. Global dynamics of cholera models with differential infectivity. *Mathematical Biosciences*. 2011;234:118–126. doi:10.1016/j.mbs.2011.09.003.

[13] Eisenberg MC, Robertson SL, Tien JH. Identifiability and estimation of multiple transmission pathways in cholera and waterborne disease. *Journal of Theoretical Biology*. 2013;324:84–102. doi:10.1016/j.jtbi.2012.12.021.

[14] Lee EC, Kelly MR Jr, Ochocki BM, et al. Model distinguishability and inference robustness in mechanisms of cholera transmission and loss of immunity. *Journal of Theoretical Biology*. 2017;420:68–81. doi:10.1016/j.jtbi.2017.01.032.

[15] Ngwa MC, Liang S, Kracalik IT, et al. Cholera in Cameroon, 2000–2012: spatial and temporal analysis at the operational (health district) and sub climate levels. *PLoS Neglected Tropical Diseases*. 2016;10:e0005105. doi:10.1371/journal.pntd.0005105.

[16] Hameni Nkwayep C, Glèlè Kakaï R, Bowong S. Prediction and control of cholera outbreak: Study case of Cameroon. *Infectious Disease Modelling*. 2024;9:892–925. doi:10.1016/j.idm.2024.04.009.

[17] Mari L, Bertuzzo E, Righetto L, Casagrandi R, Gatto M, Rodriguez-Iturbe I, Rinaldo A. Modelling cholera epidemics: the role of waterways, human mobility and sanitation. *Journal of the Royal Society Interface*. 2012;9:376–388. doi:10.1098/rsif.2011.0304.

[18] Eisenberg MC, Shuai Z, Tien JH, van den Driessche P. A cholera model in a patchy environment with water and human movement. *Mathematical Biosciences*. 2013;246:105–112. doi:10.1016/j.mbs.2013.08.003.

[19] Koelle K, Rodó X, Pascual M, Yunus M, Mostafa G. Refractory periods and climate forcing in cholera dynamics. *Nature*. 2005;436:696–700. doi:10.1038/nature03820.

[20] Posny D, Wang J. Modelling cholera in periodic environments. *Journal of Biological Dynamics*. 2014;8:1–19. doi:10.1080/17513758.2014.896482.